1. Dicliny or Unisexuality

When the flowers are unisexual only cross-pollination is possible. There are two types.

i. **Monoecious**: Male and female flowers on the same plant. Coconut, Bitter gourd. In plants like castor and maize, autogamy is prevented but geitonogamy takes place.

ii. **Dioecious**: Male and female flowers on different plants. *Borassus*, *Carica* and phoenix. Here both autogamy and geitonogamy are prevented.

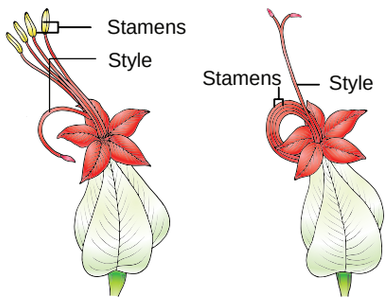
2. Monocliny or Bisexuality

Flowers are bisexual and the special adaptation of the flowers prevents self-pollination.

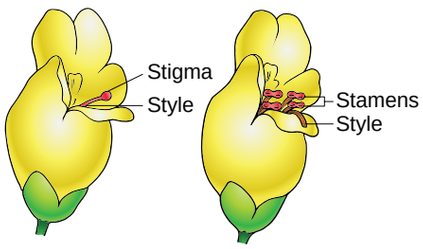
i. Dichogamy: In bisexual flowers anthers and stigmas mature at different times, thus checking self-pollination. It is of two types.

a. Protandry: The stamens mature earlier than the stigmas of the flowers. Examples: *Helianthus, Clerodendrum* (Figure 1.12 a).

b. Protogyny: The stigmas mature earlier than the stamens of the flower. Examples: *Scrophularia nodosa* and *Aristolochia bracteata* (Figure 1.12 b).



(a) Protandry - *Clerodendrum*



(b) Protogyny - *Scrophularia*

Figure 1.12 Dichogamy

ii. Herkogamy: In bisexual flowers the essential organs, the stamens and stigmas, are arranged in such a way that self-pollination becomes impossible. For example in *Gloriosa superba,* the style is reflexed away from the stamens and in *Hibiscus* the stigmas project far above the stamens (Figure 1.13).

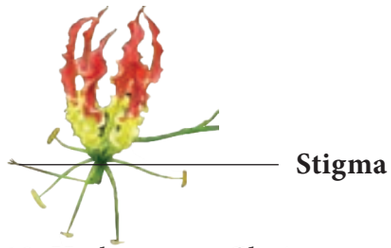
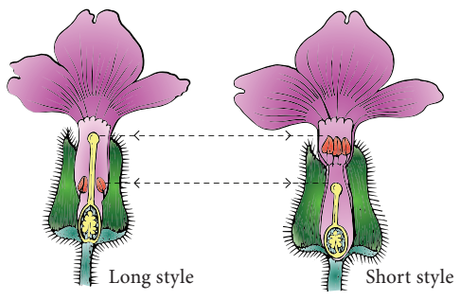


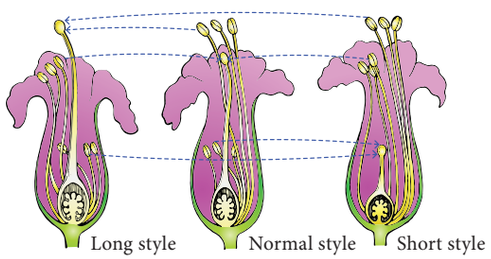
Figure 1.13 Herkogamy - *Gloriosa*

iii. Heterostyly: Some plants produce two or three different forms of flowers that are different in their length of stamens and style. Pollination will take place only between organs of the same length.(Figure 1.14)

a. Distyly: The plant produces two forms of flowers, Pin or long style, long stigmatic papillae, short stamens and small pollen grains; Thrum-eyed or short style, small stigmatic papillae, long stamens and large pollen grains. Example: *Primula* (Figure 1.14a). The stigma of the Thrum-eyed flowers and the anther of the pin lie in same level to bring out pollination. Similarly the anther of Thrum-eyed and stigma of pin ones is found in same height. This helps in effective pollination.



a) Distyly - *Primula*



b) Tristyly - *Lythrum*

Figure 1.14 Heterostyly

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